A Detailed English Exposition of

Michel Rumin, *Formes différentielles sur les variétés de contact*

J. Differential Geometry **39** (1994), 281–330

Note. This is an exposition of the mathematical content of Rumin's paper, written independently in English; it is not a translation. Statements and proofs are restated and compressed in my own words. Numbering of equations follows the original where it is useful for cross-reference. Where the scanned text of the original is garbled, this is flagged explicitly.

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1 Overall architecture

The paper has two parts and one governing idea.

The idea. On a contact manifold the natural symmetry group is not the isometry group of a Riemannian metric but the group of dilations that scale the contact distribution Q by λ and the transverse (Reeb) direction by λ^2. The de Rham complex is *isotropic*: d mixes weight-one derivatives along Q with weight-two transverse derivatives indiscriminately, so no gauge condition built from d, δ is invariant under these dilations. Rumin builds a replacement complex — differential forms "modulo contact forms" — that *is* homogeneous, proves it is a fine resolution of $\mathbb{R}$, and develops its Hodge theory.

Part I constructs the complex and proves the analytic regularity (maximal hypoellipticity) that makes Hodge theory work.

Part II develops Weitzenböck formulas using the Webster–Tanaka pseudohermitian connection and deduces vanishing theorems for $H^k(M, \mathbb{R})$, together with a pseudohermitian Myers theorem in dimension 3.

The work is Rumin's thesis, written under Gromov and Pansu. The announcement is the 1990 CRAS note (reference [10] of the paper).

Part I — A homogeneous de Rham complex on contact manifolds

2 §1. Why de Rham fails

Model. The Heisenberg group H_{2n+1}, with Lie algebra spanned by X_i, Y_i $(1 \le i \le n)$ and T, relations $[Y_i, X_i] = T$ and all other brackets zero. Put $Q = \operatorname{span}\{X_i, Y_i\}$ and let θ be the 1-form with $\theta|_Q = 0$, $\theta(T) = 1$; then $d\theta|_Q$ is symplectic, so Q is a contact structure. Equip H_{2n+1} with the left-invariant metric making $\{X_i, Y_i, T\}$ orthonormal. The dilations h_λ act on the Lie algebra by λ on Q and λ^2 on T.

For a 1-form α,

$$\delta\alpha \;=\; \sum_{i=1}^{n} \big(X_i{\cdot}\alpha(X_i) + Y_i{\cdot}\alpha(Y_i)\big) + T{\cdot}\alpha(T),$$

whereas

$$\delta(h_\lambda^*\alpha) \;=\; \lambda \sum_{i=1}^{n} \big(X_i{\cdot}\alpha(X_i) + Y_i{\cdot}\alpha(Y_i)\big) + \lambda^2\, T{\cdot}\alpha(T).$$

The two groups of terms scale differently, so co-closedness is *not* preserved by h_λ. The culprit is exactly the T-component of α.

The complementary observation: if f satisfies $df|_Q = 0$, then since $[X_1, Y_1] = T$ we get $T{\cdot}f = 0$ as well, so f is locally constant. In other words the T-component of df carries no information beyond what $df|_Q$ already contains. This is the licence to discard everything involving θ — and, because the ideal generated by θ is differential, everything involving $d\theta$ as well.

3 §2. The contact complex

Let (M^{2n+1}, Q) be a contact manifold, θ a (local) contact form, $\Omega^\bullet$ the algebra of differential forms. Define

$$I^\bullet = \{\theta \wedge \beta + d\theta \wedge \gamma : \beta, \gamma \in \Omega^\bullet\}, \qquad J^\bullet = \{\alpha \in \Omega^\bullet : \theta \wedge \alpha = 0,\ d\theta \wedge \alpha = 0\}.$$

So $I^\bullet$ is the differential ideal generated by θ and $J^\bullet$ its annihilator. (Rumin writes these with script letters; later authors often write $\mathcal{I}^\bullet$ and $\mathcal{J}^\bullet$.) Both are independent of the choice of θ within its conformal class $\theta \mapsto f\theta$, and both are d-stable. Hence d descends to $\Omega^\bullet/I^\bullet$ and restricts to $J^\bullet$; write d_Q for the induced operators.

The Lefschetz obstruction. Since $d\theta|_Q$ is symplectic, the hard Lefschetz algebra (Weil) says that

$$L = d\theta \wedge \cdot \ : \ \Lambda^k Q^* \longrightarrow \Lambda^{k+2} Q^*$$

is injective for $k \leq n-1$ and surjective for $k \geq n-1$. Consequently

$$\Omega^k/I^k = 0 \quad (k \geq n+1), \qquad J^k = 0 \quad (k \leq n).$$

Each induced complex is only *half* a complex. They must be spliced, and the splice cannot be first order.

Proposition 1 (the complex)**.** *There is a second-order linear differential operator $D : \Omega^n/I^n \to J^{n+1}$ such that*

$$0 \to \mathbb{R} \to C^\infty(M) \xrightarrow{d_Q} \Omega^1/I^1 \xrightarrow{d_Q} \cdots \xrightarrow{d_Q} \Omega^n/I^n \xrightarrow{D} J^{n+1} \xrightarrow{d_Q} \cdots \xrightarrow{d_Q} J^{2n+1} \to 0$$

is a resolution of the constants. Its cohomology is therefore $H^\bullet_{dR}(M)$.

Construction of D. Identify $\Omega^n/\{\theta \wedge \alpha : \alpha \in \Omega^{n-1}\} \cong \Lambda^n Q^*$.

Lemma 2. *For $\alpha \in \Lambda^n Q^*$ there is a* unique *lift $\tilde\alpha \in \Omega^n$ of α with $d\tilde\alpha \in J^{n+1}$. Set $\tilde D\alpha := d\tilde\alpha$.*

Proof sketch. Take any lift $\bar\alpha$ and look for $\beta \in \Lambda^{n-1}Q^*$ such that $\tilde\alpha = \bar\alpha + \theta \wedge \beta$ satisfies $\theta \wedge d\tilde\alpha = 0$, i.e. $(d\bar\alpha + d\theta \wedge \beta)|_Q = 0$. This determines β uniquely because L is an *isomorphism* in degree $n-1$ — precisely the degree where injectivity and surjectivity overlap. The remaining condition is automatic: $d\theta \wedge d\tilde\alpha = d(\theta \wedge d\tilde\alpha) = 0$. □

One then checks that $\tilde D$ descends to $\Omega^n/I^n \cong \Lambda^n Q^*/\{\overline{d\theta \wedge \beta} : \beta \in \Omega^{n-2}\}$: for $\beta \in \Omega^{n-2}$ the form $d\theta \wedge \beta - \theta \wedge d\beta$ is closed, hence is the distinguished lift of $\overline{d\theta \wedge \beta}$, and $\tilde D$ annihilates it.

Thus D is **second order in horizontal derivatives**: one derivative is spent solving for the lift, one on differentiating it. The operators d_Q are first order horizontal. Crucially, D and all the d_Q depend only on the contact structure Q, not on the choice of θ — the complex is a contact invariant.

Local exactness

Four cases, all running on the same trick: modify a lift by $\theta\wedge\gamma$ and then use injectivity or surjectivity of L in the appropriate degree.

1. $k \le n-1$. If $d_Q\alpha = 0$, a lift satisfies $d\tilde\alpha = \theta\wedge\beta - d\theta\wedge\gamma$; replacing $\tilde\alpha$ by $\tilde\alpha - \theta\wedge\gamma$ kills the second term, so $d\tilde\alpha = \theta\wedge\beta$. Differentiating, $0 = d\theta\wedge\beta + \theta\wedge d\beta$, and restricting to Q gives $L(\beta|_Q) = 0$, hence $\beta|_Q = 0$ by injectivity. So $\beta = \theta\wedge\delta$ and $d\tilde\alpha = 0$; locally $\tilde\alpha = d\mu$, and projecting gives $\alpha = d_Q\bar\mu$.

2. $k \ge n+2$. If $\alpha \in J^k$ is closed, write $\alpha = d\beta$ locally and use *surjectivity* of L in degree $k-3 \ge n-1$ to write $\beta = \theta\wedge\gamma + d\theta\wedge\mu$. Then $\beta' := \beta - d(\theta\wedge\mu) = \theta\wedge(\gamma + d\mu)$ satisfies $d\beta' = \alpha$, $\theta\wedge\beta' = 0$ and $d\theta\wedge\beta' = \theta\wedge\alpha = 0$, so $\beta' \in J^{k-1}$.

3. *Degree n.* $D\alpha = 0$ means the distinguished lift is closed; take a primitive and project.

4. *Degree $n+1$.* If $\alpha \in J^{n+1}$ is closed then $\alpha = d\beta$ with $\beta \in \Omega^n$, and $\alpha = D(\bar\beta)$.

4 §3. Hypoelliptic regularity

4.1 Setting up the metric

Fix θ and let T be the Reeb field ($\theta(T) = 1$, $\mathcal{L}_T\theta = 0$). Choose a *calibrated* complex structure J on Q:

$$J^2 = -\operatorname{Id}, \qquad d\theta(X, JY) = -d\theta(JX, Y), \qquad d\theta(X, JX) > 0 \ \ (X \neq 0).$$

Existence (after Weinstein): write $d\theta|_Q = g|_Q(A\cdot,\cdot)$ with A antisymmetric and polar-decompose $A = S\cdot J$ with S symmetric positive definite and J orthogonal.

Set $g_Q = d\theta(\cdot, J\cdot)$ on Q and the *adapted metric* $g_\theta = \theta^2 + d\theta(\cdot, J\cdot)$ on TM (extending J by $JT = T$).

The quotients Ω^k/I^k ($k \le n$) are then identified with genuine subspaces $E^k \subset \Omega^k Q$, the **primitive horizontal forms**:

$$i_T\alpha = 0 \quad \text{and} \quad \Lambda\alpha = 0,$$

where Λ is the adjoint of $L = d\theta\wedge\cdot$. (Rumin writes these with a script letter; modern papers write E^k or $\mathcal{R}^k$.)

The Hodge star, defined by $\alpha\wedge *\beta = (\alpha,\beta)\,\theta\wedge(d\theta)^n$, puts E^k and E^{2n+1-k} in duality. Adjoints:

$$\delta_Q = (-1)^k * d_Q * \ \text{ on } E^k \ (k \neq n+1), \qquad D^* = (-1)^{n+1} * D * \ \text{ in degree } n+1.$$

An integration-by-parts argument verifies that these are the genuine formal adjoints; it uses that $d_Q = d$ in degrees $> n$, that $\operatorname{Im}(d_Q - d) \subset I^\bullet$ in degrees $< n$, and that I-forms annihilate J-forms under exterior product.

Remark 3 (the payoff of §1). δ_Q is *conformally invariant* under $g_\theta \mapsto g_{k\theta} = k^2\theta^2 + k\,d\theta(\cdot, J\cdot)$ for constant k. This is exactly the homogeneity that the de Rham codifferential lacked.

4.2 The Laplacians and maximal hypoellipticity

The complex is **not elliptic**: on functions on H_{2n+1} one computes $(d_Q\delta_Q + \delta_Q d_Q)f = -\sum_i (X_i^2 + Y_i^2)f$, the sub-Laplacian. The right regularity notion is weaker.

Definition 4 (Helffer–Nourrigat)**.** P is *maximally hypoelliptic* if locally

$$\|f\|_{2,k,Q} \;\le\; K\big(\|Pf\|_2 + \|f\|_2\big), \qquad \|f\|_{2,k,Q} = \sum_{l\le k} \big\|X_{i_1}X_{i_2}\cdots X_{i_l} f\big\|_2,$$

with $\{X_1,\dots,X_{2n}\}$ a frame for Q and K independent of f (compactly supported in the neighbourhood under study). "Maximal" means P controls the largest possible number of horizontal derivatives.

Theorem 5. *On a contact manifold with an adapted metric, the following are maximally hypoelliptic:*

degree	*operator*
$0 \le k \le n-1$	$\Delta_Q = (n-k)\, d_Q\delta_Q + (n-k+1)\, \delta_Q d_Q$ *on* E^k
$k = n$	$\Delta_Q = (d_Q\delta_Q)^2 + D^*D$
$k = n+1$	$\Delta_Q = DD^* + (\delta_Q d_Q)^2$
$n+2 \le k \le 2n+1$	$\Delta_Q = (n-k+1)\, d_Q\delta_Q + (n-k)\, \delta_Q d_Q$

Remark 6 (normalization — important). Note the degree-dependent coefficients $(n-k)$, $(n-k+1)$. This is Rumin's own normalization and it is *not* the convention used in much of the later literature, where the Laplacians are often defined by squaring uniformly, giving order 4 in every degree. In Rumin's convention the operator is order 2 in Q-derivatives away from the middle and order 4 in the middle degrees $k = n, n+1$.

Corollary 7 (Hodge theory)**.** *Weak solutions of* $\Delta_Q\alpha = \beta$ *with* β *smooth are smooth;* $E^\bullet$ *(compact supports) decomposes orthogonally as* $\ker\Delta_Q \oplus \operatorname{Im}\Delta_Q$ *with* $\ker\Delta_Q$ *finite dimensional; each* d_Q*-cohomology class with compact support has a unique co-closed representative; and on compact* M *the cohomology is represented by* Δ_Q*-harmonic forms.*

This uses the Folland–Stein–Kohn compact embedding and regularity theorems for the spaces $L^2_{k,Q}$.

4.3 Proof strategy: compare with scalar operators

Rumin's route is to deduce hypoellipticity of Δ_Q from that of simpler, nearly scalar operators.

The horizontal differential. On $\Omega^\bullet Q := \{\alpha : i_T\alpha = 0\}$ define $d_H = \Pi\circ d$, with Π the orthogonal projection onto $\Omega^\bullet Q$. It is first order horizontal, and

$$d_H^2 = -L\,\mathcal{L}_T = -\mathcal{L}_T\,L. \tag{1}$$

(Proof: $d_H\alpha = d\alpha - \theta\wedge\mathcal{L}_T\alpha$; differentiate and project, using $i_T(d\theta\wedge\mathcal{L}_T\alpha) = 0$.)

Orders and the $o(k)$ calculus. The *order* of an operator on $\Omega^\bullet Q$ counts horizontal derivatives only. Because $T\cdot f = (Y_1X_1 - X_1Y_1)f + X\cdot f$ for a suitable $X \in Q$, the transverse derivative ∂_T has **order 2**. Write $o(k)$ for any operator of order $< k$, and $A \simeq B$ for $A = B + o(p)$ where p is their common order. This bookkeeping is the technical engine of the whole section.

Bidegree. J splits $\Omega^k Q \otimes \mathbb{C} = \bigoplus_{p+q=k} \Omega^{p,q} Q$. Write $\partial_H := d_H^{1,0}$, $\bar\partial_H := d_H^{0,1}$. Then

$$d_H = \partial_H + \bar\partial_H + T(J), \qquad d_H = \partial_H + \bar\partial_H + o(1), \tag{2, 2'}$$

where $T(J)$ is a zeroth-order tensor vanishing when J is integrable; likewise $\mathcal{L}_T = (\mathcal{L}_T)^{0,0} + o(1)$, the $o(1)$ vanishing when the Reeb flow preserves J. Comparing (1), (2′) and (4) yields

$$\partial_H^2 = o(2), \qquad \bar\partial_H^2 = o(2), \qquad \partial_H \bar\partial_H + \bar\partial_H \partial_H = -L\,\mathcal{L}_T + o(1). \tag{5, 6}$$

With $*_Q : \Omega^\bullet Q \to \Omega^{2n-\bullet} Q$ defined by $\alpha \wedge *_Q \beta = (\alpha, \beta)\, d\theta^n$ one obtains $\delta_H = -*_Q d_H *_Q$; setting $d_H^J = J^{-1} d_H J$, the **Kähler identities hold up to** $o(1)$:

$$[\Lambda, d_H] = -\delta_H^J + o(1), \qquad [\Lambda, d_H^J] = \delta_H + o(1). \tag{8}$$

The proof compares local expressions in an orthonormal frame $\{X_i, JX_i\}$ with the standard Kähler ones on $\mathbb{C}^n$: the algebraic operators $J, L, \Lambda, *_Q$ are literally identical, and Cartan's formula shows d_H differs only by algebraic terms built from the brackets of the X_i, JX_i — hence by lower order.

The key computations. Set $Z_k = (X_k - iY_k)/\sqrt{2} \in Q^{1,0}$, $Z_{\bar k} = (X_k + iY_k)/\sqrt{2} \in Q^{0,1}$, with dual coframe $\theta^k, \theta^{\bar k}$, and the usual operators $i_k\alpha = \alpha(Z_k, \cdot)$, $e_k\alpha = \theta^k \wedge \alpha$, $\partial_k \alpha = \sum (Z_k \cdot \alpha_{I,J}) \theta^I \wedge \theta^{\bar J}$. Then

$$\partial_{\bar k}\partial_k - \partial_k \partial_{\bar k} \simeq i\partial_T, \quad L = i\sum_k e_k e_{\bar k}, \quad \Lambda = i\sum_k i_k i_{\bar k}, \quad \partial_H \simeq \sum \partial_k e_k, \quad \partial_H^* \simeq -\sum \partial_{\bar k} i_k, \quad \mathcal{L}_T \simeq \partial_T.$$

Proposition 8. *On $\Omega^k Q$,*

$$\Delta_{\partial_H} - \Delta_{\bar\partial_H} = i(k-n)\mathcal{L}_T + o(2),$$

and $\Delta_H = \Delta_H^{0,0} + o(2)$; that is, Δ_H preserves bidegree up to lower order.

Proposition 9. *On $\Omega^{p,q} Q$,*

$$\Delta_{\partial_H} \simeq -\tfrac{1}{2}\sum_k (X_k^2 + Y_k^2) + i\Big(p - \tfrac{n}{2}\Big)\partial_T, \qquad \Delta_H \simeq -\sum_k (X_k^2 + Y_k^2) + i(p-q)\partial_T.$$

These are the Folland–Stein operators $-\sum(X_k^2 + Y_k^2) + i\lambda T$, which are hypoelliptic (Kohn) and maximally so (Folland–Stein) precisely when

$$\boxed{|\lambda| \notin \{n,\, n+2,\, n+4, \dots\}.}$$

Corollary 10. *Δ_H is maximally hypoelliptic on $\Omega^{p,q}$ for $p, q < n$.*

Proposition 11 (relating d_Q to d_H)**.** *On E^k, $0 \le k \le n-1$,*

$$d_Q \simeq d_H + (n-k+1)^{-1} L \delta_H^J, \qquad \delta_Q = \delta_H; \tag{9}$$

and in degree n,

$$i_T D \simeq \mathcal{L}_T - d_H \delta_H^J. \tag{10}$$

Proof sketch of (9). $d_Q\alpha$ is the orthogonal projection of $d_H\alpha$ onto $\ker\Lambda$, so by Weil's lemma $d_Q\alpha = \sum_{s\ge 0} a_s L^s \Lambda^s d_H \alpha$ with universal coefficients a_s. The series truncates: by (8), $\Lambda d_H \alpha = -\delta_H^J \alpha + o(1)\alpha$ for $\alpha \in E^k \subset \ker\Lambda$, and $[\Lambda, \delta_H^J] = 0$, so $\Lambda^s d_H \alpha = o(1)\alpha$ for $s \ge 2$. To find a_1, apply Λ to $d_Q\alpha$ and use $[\Lambda, L] = (n-k+1)\,\mathrm{Id}$ in degree $k-1$; identifying degree-one terms gives $a_1 = -(n-k+1)^{-1}$. Finally $\delta_Q = \delta_H$ on E^k because $[\delta_H, \Lambda] = 0$. □

Proposition 12. *$\Delta_Q \simeq \Delta_Q^{0,0}$: the contact Laplacian* almost *preserves bidegree.*

The comparison device. For linear differential operators A, B on a vector bundle over M, write $A \gtrsim B$ if $A \simeq B + P^*P$ for some linear differential operator P.

Proposition 13. *If $A \gtrsim B \gtrsim 0$ then maximal hypoellipticity of B implies that of A.*

This rests on the **Helffer–Nourrigat criterion**. To $P = \sum_{|\alpha|\le k} a_\alpha(x) X^\alpha$ attach at each x_0 the frozen left-invariant model P_{x_0} and its top homogeneous part $P^k_{x_0} = \sum_{|\alpha|=k} a_\alpha(x_0) X^\alpha$; for each irreducible unitary representation Π of H^{2n+1} let $\Pi(P)$ be the image of $P^k_{x_0}$ acting on the smooth vectors, identified with $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$. Concretely Π is equivalent either to $\Pi_{a,b}(X_k) = ia_k$, $\Pi_{a,b}(Y_k) = ib_k$, $\Pi_{a,b}(T) = 0$ $(a, b \in \mathbb{R}^n)$, or to $\Pi_\lambda(X_k) = \partial/\partial x_k$, $\Pi_\lambda(Y_k) = i\lambda x_k$, $\Pi_\lambda(T) = i\lambda$ $(\lambda \in \mathbb{R}^*)$. Then:

$$P \text{ is maximally hypoelliptic at } x_0 \iff \Pi(P) \text{ is injective for every nontrivial } \Pi.$$

Since $\Pi(A) = \Pi(B)$ when $A \simeq B$, $\Pi(AB) = \Pi(A)\Pi(B)$ and $\Pi(A^*) = \Pi(A)^*$, the proposition follows at once. Darboux's theorem reduces everything to the Heisenberg group.

Remark 14. This is the **Rockland condition** in Helffer–Nourrigat form, and it is the exact analytic input of the paper.

Illustration. With $\Delta_K f = -\sum_{i=1}^n (X_i^2 + Y_i^2) f$ the Kohn Laplacian acting componentwise on forms, one has

$$n\Delta_H \gtrsim \Delta_K, \tag{13}$$

via the explicit $P : \Omega^{p,q}Q \to \Lambda^{1,0}Q \otimes \Omega^{p,q}Q + \Lambda^{0,1}Q \otimes \Omega^{p,q}Q$, $\alpha \mapsto (\lambda_{1,0}\partial\alpha, \lambda_{0,1}\bar\partial\alpha)$, with $\lambda^2_{1,0} = n - 1 + q - p$ and $\lambda^2_{0,1} = n - 1 + p - q$.

Proposition 15 (key inequalities)**.**

(i) $(n-k+2)\Delta_Q \gtrsim (n-k)(n-k+1)\Delta_H$ *on* E^k, $k \le n-1$;

(ii) $4\Delta_Q \gtrsim \Delta_H^2$ *on* $E^{p,q}$ *with* $p+q = n$, $p, q < n$;

(iii) $\Delta_Q \simeq (\Delta_K + i(n+1)\partial_T)^2$ *on* $E^{n,0}$, *and* $(\Delta_K - i(n+1)\partial_T)^2$ *on* $E^{0,n}$;

(iv) the formulas are valid in complementary degree upon replacing k by $2n+1-k$.

The bidegrees $(n, 0)$ and $(0, n)$ are the delicate ones: $P_{n+1} := \Delta_K + i(n+1)\partial_T$ is no longer positive, but remains hypoelliptic (Folland–Rothschild–Stein). Rumin verifies this directly by the Helffer–Nourrigat criterion, using the intertwining $\partial_{\bar k} P_{n+1} \simeq P_{n-1}\partial_{\bar k}$ and the fact that $P_{n-1} \gtrsim \Delta_K/n$ is hypoelliptic: if $\Pi(P_{n+1})f = 0$ then $\Pi(P_{n-1})\Pi(\partial_{\bar k})f = 0$, whence $\Pi(\partial_{\bar k})f = 0$; writing $P_{n+1} \simeq n^{-1}\sum \partial_{\bar k}\partial_k - (2 + n^{-1})\sum \partial_k\partial_{\bar k}$ with $-\partial_k \simeq \partial_k^*$ gives $\Pi(\partial_k)f = 0$, so $\Pi(X)f = 0$ for all $X \in Q$ and $f = 0$.

Proposition 16 (square roots)**.** *The operators*

$$P^\pm = d_Q\delta_Q \pm i^{n+1} * D$$

are self-adjoint and satisfy $\Delta_Q = (P^+)^2 = (P^-)^2$ in the middle degree (a consequence of $D^* = (-1)^{n+1} * D *$ and $*^2 = 1$). They do not preserve bidegrees, only the pairs $E^{p,q} \oplus E^{p+1,q-1}$; more precisely, P^+ (resp. P^-) preserves, up to o,

$$E^{\lambda+2k,\, n-\lambda-2k} \oplus E^{\lambda+2k+1,\, n-\lambda-2k-1}, \qquad k \in \mathbb{Z},$$

with $\lambda = (n+1)(n+2)/2 + 1$ (resp. $(n+1)(n+2)/2$).

Since $E^{0,n+1} = E^{n+1,0} = 0$, the extreme bidegrees *are* preserved — which explains the perfect-square formulas in (iii) above.

Part II — Contact complex and curvature

5 §1. The adapted connection

We now look for a connection with the same homogeneity as Δ_Q: invariant under $g_\theta \mapsto g_{k\theta}$, i.e. $\nabla g = 0$ and $\nabla\theta = 0$.

Levi-Civita fails. If $\nabla\theta = 0$ then $\nabla_X Y \in Q$ for $X, Y \in Q$, and on $Q \times Q$,

$$\theta(\mathrm{Tor}(X,Y)) = \theta(\nabla_X Y - \nabla_Y X - [X,Y]) = d\theta(X,Y) \neq 0,$$

so torsion is forced. For any g_θ-metric connection,

$$2(\nabla_X\theta)(Y) = d\theta(X,Y) - \theta(\mathrm{Tor}(X,Y)) - (\mathrm{Tor}(T,X),Y) - (\mathrm{Tor}(T,Y),X) - (J(\mathcal{L}_T J)(Y),X). \quad (*)$$

Since $(J(\mathcal{L}_T J)(X),Y) = (J(\mathcal{L}_T J)(Y),X)$, splitting $(*)$ into symmetric and antisymmetric parts gives: $\nabla\theta = 0$ if and only if

$$\theta(\mathrm{Tor}(X,Y)) = d\theta(X,Y) \quad \text{and} \quad X \mapsto \mathrm{Tor}(T,X) + \tfrac{1}{2}J(\mathcal{L}_T J)(X) \text{ is antisymmetric for } g_\theta.$$

Following Webster, Tanaka and Tanno one chooses

$$\mathrm{Tor}(X,Y) = d\theta(X,Y)\,T \quad (X,Y \in Q), \qquad \mathrm{Tor}(T,X) = -\tfrac{1}{2}J(\mathcal{L}_T J)(X).$$

When J is integrable ($[Q^{1,0}, Q^{1,0}] \subset Q^{1,0}$) this connection satisfies $\nabla J = 0$; by Webster it is the unique metric connection preserving θ and J whose torsion satisfies $\mathrm{Tor}(T, JX) = -J\,\mathrm{Tor}(T,X)$, i.e. $\mathrm{Tor}(T, Q^{0,1}) \subset Q^{1,0}$. From here on J is assumed integrable and the data $(\theta, J, g = \theta^2 + d\theta(\cdot, J\cdot))$ is called a **pseudohermitian structure** — the geometry of a strictly pseudoconvex hypersurface in $\mathbb{C}^n$.

Two curvature quantities carry the whole story.

- $R_1 X := \mathrm{Tor}(T,X) = -\frac{1}{2}J(\mathcal{L}_T J)(X)$, the *pseudohermitian torsion*, measuring how far the Reeb flow is from preserving J and g. One has $R_1 = 0$ iff the Reeb flow is Riemannian.
- R_2, a "Ricci curvature along Q". On 1-forms,

$$\frac{2n}{n-1}(R_2\alpha,\alpha) = \mathrm{Ricci}_Q(\alpha^*,\alpha^*) + \mathrm{Ricci}_Q(J\alpha^*, J\alpha^*), \quad (27)$$

  where $\alpha^* \in Q$ is the dual vector.

Remark 17 (the sign phenomenon that drives everything). $R_2 J = J R_2$ but $R_1 J = -J R_1$, so $(R_1 J\alpha, J\alpha) = -(R_1\alpha,\alpha)$. Thus R_1 is a **"signless" quadratic form**, which is why it can only ever be *dominated*, never used as a positivity hypothesis.

6 §2. Pseudohermitian vanishing theorems

With J integrable the o's of Part I become explicit: now $d_H = \partial_H + \bar\partial_H$ exactly, and

Lemma 18 (exact Kähler identities)**.**

$$[\Lambda, d_H] = -\delta_H^J, \qquad [\Lambda, d_H^J] = \delta_H. \tag{17}$$

The proof is at a point m, using a *normal frame*: start from an orthonormal basis $\{X_1, JX_1, \dots, X_n, JX_n\}$ of Q_m and extend by parallel transport along geodesics from m. The resulting frame $\{X_i, Y_i = JX_i\}$ is orthonormal in Q (the pseudohermitian connection is metric and preserves θ, J), and $\nabla_X Y = 0$ at m, so $[X,Y] = -d\theta(X,Y)T$: the brackets are collinear with T and the Kähler computations go through verbatim.

The bidegree-violating parts are then identified. From (16) and $[\Lambda, L] = n-k$, for $\alpha \in E^k = \Omega^k Q \cap \ker\Lambda$,

$$\big((\Delta_H - \Delta_H^{0,0})\alpha, \alpha\big) = (n-k)(R_1\alpha, \alpha), \qquad \big((\Delta_Q - \Delta_Q^{0,0})\alpha, \alpha\big) = (n-k)(n-k+1)(R_1\alpha, \alpha), \tag{18, 21}$$

where, explicitly,

$$R_1\alpha(X_1,\dots,X_k) = -\sum_{i=1}^k \alpha\big(X_1,\dots,J\,\mathrm{Tor}(T,X_i),\dots,X_k\big) = -\tfrac12\sum_{i=1}^k \alpha\big(X_1,\dots,(\mathcal{L}_T J)(X_i),\dots,X_k\big). \tag{19}$$

The bidegree-preserving part is

$$\Big(\frac{n-k+2}{n-k+1}\Big)\Delta_Q^{0,0} = (n-k)\Delta_H^{0,0} + 2(d_Q^{1,0})^* d_Q^{1,0} + 2(d_Q^{0,1})^* d_Q^{0,1}. \tag{22}$$

Lemma 19 (Weitzenböck–Tanaka)**.** *On $\Omega^{p,q}Q$,*

$$\Delta_H^{0,0} = \Big(1 + \frac{p-q}{n}\Big)(\nabla^{0,1})^*\nabla^{0,1} + \Big(1 - \frac{p-q}{n}\Big)(\nabla^{1,0})^*\nabla^{1,0} + R_2, \tag{23}$$

where R_2 is an algebraic "trace" of the pseudohermitian curvature.

Tanaka proved this when $\mathcal{L}_T J = 0$; the proof is formally identical here. It uses a normal frame, the identity $\partial_H \equiv \sum e_k \nabla_k$, $\partial_H^* \equiv -\sum i_k \nabla_{\bar k}$ near m, and the commutation relation

$$\nabla_{\bar k}\nabla_l - \nabla_l\nabla_{\bar k} - \delta_{kl}\nabla_{iT} = R(Z_{\bar k}, Z_l), \tag{24}$$

coming from $[Z_{\bar k}, Z_l] = i\delta_{kl}T$, together with

$$n\nabla_{iT} \equiv -(\nabla^{1,0})^*\nabla^{1,0} + (\nabla^{0,1})^*\nabla^{0,1} - \sum_k R(\bar k, k). \tag{25}$$

The resulting expression for R_2 can be rewritten as

$$R_2 = \sum_{k,l}\big(e_k R(\bar l, k) i_l + e_{\bar k} R(l, \bar k) i_{\bar l}\big) + \Big(1 + \frac{q-p}{n}\Big)R^{1,0} - \Big(1 - \frac{q-p}{n}\Big)R^{0,1}, \tag{26}$$

which makes self-adjointness manifest.

Theorem 20 (vanishing criterion, $k \neq n, n+1$)**.** *$H^k(M,\mathbb{R}) = 0$ for $k < n$ provided that for all $\alpha \in E^k \setminus \{0\}$,*

$$\big((R_2 + (n-k+2)R_1)\alpha, \alpha\big) > 0,$$

equivalently

$$(R_2\alpha, \alpha) > (n-k+2)\,\big|(R_1\alpha, \alpha)\big|$$

— that is, R_2 must dominate R_1.

The proof is immediate from (21), (22) and (23) applied to a Δ_Q-harmonic form; the two forms of the condition are equivalent because $R_2 J = J R_2$ and $R_1 J = -J R_1$.

Remark 21 (no general criterion in middle degree). In degrees n and $n+1$ the Laplacians are fourth order, while the Weitzenböck machinery above is second order. Rumin states plainly that he does not know how to obtain a general criterion there.

Orders as an explanatory principle (§5). Under $g_\theta \mapsto g_{k^2\theta}$,

$$\|R_2\|,\ \|R_1\|,\ \|\operatorname{Tr}_Q(\nabla\cdot R_1)\|,\ \|\nabla_T R_1\| \ \longmapsto\ \lambda^{-2}\|R_2\|,\ \lambda^{-2}\|R_1\|,\ \lambda^{-3}\|\operatorname{Tr}_Q(\nabla\cdot R_1)\|,\ \lambda^{-4}\|\nabla_T R_1\|,$$

so these curvature quantities have weights $2, 2, 3, 4$. The contact Laplacian is order 2 in Q; the Hodge–de Rham Laplacian has weight 4 because it takes two derivatives along the transverse direction T, itself of order 2. Hence the contact criterion involves only curvature of weight ≤ 2, while the Riemannian Ricci criterion involves weight 4. This is the paper's own explanation of why the contact criterion is sharper.

7 §3. Vanishing torsion: transversally Kähler flows

Suppose $R_1 = 0$, so the Reeb flow preserves J, g, ∇ and its curvature. Assume further that all orbits are closed with the same period. Then $N = M/\langle T\rangle$ is a Kähler manifold and $\pi : M \to N$ a circle bundle.

Proposition 22. *For $k \leq n$,*

$$\{\alpha \in E^k : \Delta_Q\alpha = 0\} = \{\alpha \in \Omega^k M : \Delta\alpha = 0\} = \pi^*\{\alpha \in \Lambda^k N : \Delta\alpha = \Lambda\alpha = 0\}.$$

That is, the Δ_Q-harmonic forms are exactly the pullbacks of the primitive *harmonic forms of N.*

The proof uses

$$d_H = \operatorname{proj}_{\ker i_T} d = d - \theta\wedge i_T d = d - \theta\wedge\mathcal{L}_T, \qquad \delta_H = \operatorname{proj}_{\ker i_T}\delta = \delta - \theta\wedge\Lambda, \tag{28, 29}$$

the second because $i_T\delta + \delta i_T = \Lambda$ is adjoint to $d(\theta\wedge\cdot) + \theta\wedge d = L$. For $k < n$ one shows $\Delta_Q\alpha = 0$ forces $\Delta_H\alpha = 0$, hence each bidegree component is harmonic for ∂_H and $\bar\partial_H$ separately, hence $\mathcal{L}_T\alpha = 0$ by Proposition 1 of Part I; then (28)–(29) give $d\alpha = \delta\alpha = 0$.

In degree n the argument is prettier: harmonic forms are invariant under Riemannian flows, and one shows directly

$$\mathcal{L}_T\alpha = i_T D\alpha + d_H\delta_H^J\alpha = d_Q\delta_Q^J\alpha,$$

so $\mathcal{L}_T\alpha$ is the harmonic representative of the zero d_Q-cohomology class, hence $\mathcal{L}_T\alpha = 0$.

Consequence. $H^1(M,\mathbb{R}) = H^1(N,\mathbb{R})$, and since ∇^H is the pullback of the Levi-Civita connection of N (it is the unique connection preserving $g|_Q, \theta$ with $\mathrm{Tor}(T,X) = \mathcal{L}_T JX = 0$ here), one has $\pi R^H(X,Y)Z = R^N(\pi X, \pi Y)\pi Z$, so positivity of R_2 is exactly positivity of Ricci^N. The contact criterion therefore **recovers** the classical Riemannian one in this limit.

8 §4. H^1 in dimension 3

Here $\Delta_Q = (d_Q\delta_Q)^2 + D^*D$ is a priori fourth order, so the previous method does not apply directly. The escape is the square root from Proposition 8 of Part I:

$$P = d_Q\delta_Q - *D, \qquad P^* = P, \qquad \Delta_Q = P^2.$$

Expressing P via the pseudohermitian connection, using (10) and (11),

$$P = d_Q\delta_Q + J(i_T D) = J\mathcal{L}_T + \Delta_H^{0,0},$$

and since $\Delta_H^{0,0} = 2(\nabla^{0,1})^*\nabla^{0,1}$ on $\Omega^{1,0}$ and $2(\nabla^{1,0})^*\nabla^{1,0}$ on $\Omega^{0,1}$, one obtains the Weitzenböck formula

$$P = 2\nabla_-^*\nabla_- + J\nabla_T - R_1, \qquad \begin{aligned}\nabla_+\alpha &= \nabla^{1,0}\alpha^{1,0} + \nabla^{0,1}\alpha^{0,1},\\ \nabla_-\alpha &= \nabla^{1,0}\alpha^{0,1} + \nabla^{0,1}\alpha^{1,0}.\end{aligned}$$

Using $J\nabla_T = -\nabla_+^*\nabla_+ + \nabla_-^*\nabla_- - R_2$ (from (25)) one gets a **one-parameter family** of Weitzenböck formulas

$$P = (2-\lambda)\nabla_-^*\nabla_- + \lambda\nabla_+^*\nabla_+ + (1+\lambda)J\nabla_T + \lambda R_2 - R_1, \qquad \lambda \in \mathbb{R}. \tag{30_λ}$$

The quantity $(J\nabla_T\alpha, \alpha)$ has no sign, so no criterion drops out immediately: P is a "signless" operator. The repair goes as follows. For α harmonic, $i_T D\alpha = 0$ gives $\mathcal{L}_T\alpha = d_Q(\delta_H^J\alpha)$, whence

$$|\mathcal{L}_T\alpha|^2 = \big(\delta_Q d_Q(\delta_H^J\alpha), \delta_H^J\alpha\big) \le |\delta_H^J\alpha|\,|\delta_Q\mathcal{L}_T\alpha|, \qquad \delta_Q\mathcal{L}_T\alpha = [\delta_Q, \mathcal{L}_T + \mathcal{L}_T^*]\alpha = 2\delta_Q JR_1\alpha,$$

using $\mathcal{L}_T = \nabla_T + JR_1$, hence $\mathcal{L}_T^* = -\nabla_T - R_1 J = -\nabla_T + JR_1$ and $\mathcal{L}_T + \mathcal{L}_T^* = 2JR_1$. With $|\delta_Q\beta| \le |\nabla_Q\beta|$ and $\nabla J = 0$ this yields

$$|\mathcal{L}_T\alpha|^2 \le 2|\nabla_Q\alpha|\big(|R_1||\nabla_Q\alpha| + |\nabla_Q R_1||\alpha|\big),$$

and then, without optimizing,

$$|\nabla_T\alpha| \;\le\; |R_1||\alpha| + \sqrt{2}\Big(|R_1|^{1/2}|\nabla_Q\alpha| + \big(|\nabla_Q R_1||\nabla_Q\alpha||\alpha|\big)^{1/2}\Big).$$

Substituting into (30_1), normalizing $|\alpha| = 1$, and forcing the resulting quadratic in $|\nabla_Q\alpha|$ to have negative discriminant gives:

Theorem 23. *If for every unit $\alpha \in \Omega^1 Q$,*

$$(R_2\alpha, \alpha) > 7\,\|\mathcal{L}_T J\| + (1+\sqrt{2})\,\|\nabla_Q\mathcal{L}_T J\|^{2/3},$$

then $H^1(M,\mathbb{R}) = 0$.

The exponents are forced by scale invariance: under $g_\theta \mapsto g_{k\theta}$, $|R_1| \mapsto |R_1|/k$, $R_2 \mapsto R_2/k$, $|\nabla_Q R_1| \mapsto |\nabla_Q R_1|/k^{3/2}$, and $|\nabla\alpha|^2 \mapsto |\nabla\alpha|^2/k$ for unit α.

Remark 24. This criterion needs a **weight-3** curvature term $\nabla_Q R_1$ — worse than the ≤ 2 of the higher-dimensional criterion, but still better than the weight-4 Riemannian condition. The reason is structural: to conclude in dimension 3 one had to bring in the third-order operators $\delta_Q\mathcal{L}_T$ and $\mathcal{L}_T\delta_Q$.

9 §5. Riemannian versus pseudohermitian curvature

To compare with the classical Ricci criterion, work with $g_\lambda = \lambda\theta^2 + d\theta(\cdot, J\cdot)$ (note $\mathrm{Ricci}^L(g_{\lambda\theta}) = \mathrm{Ricci}^L(\lambda g_\lambda) = \lambda\,\mathrm{Ricci}^L(g_\lambda)$, so nothing is lost), and set

$$D(X,Y) = \nabla^L_X Y - \nabla_X Y.$$

Since both connections preserve g_λ, the Koszul formula gives $2(D(X,Y),Z) = (\mathrm{Tor}\,\nabla(Y,X),Z) + (\mathrm{Tor}\,\nabla(X,Z),Y) + (\mathrm{Tor}\,\nabla(Y,Z),X)$, and distinguishing cases:

$$2\lambda D(X,Y) = \big(\lambda d\theta(Y,X) - 2(\mathrm{Tor}(T,X),Y)_1\big)T \ \ (X,Y\in Q), \quad 2D(T,Y) = \lambda JY, \quad 2D(X,T) = 2\,\mathrm{Tor}(T,X) + \lambda JX,$$

Sectional curvatures of ∇^L then split into three types — a horizontal-generic plane (X,Y) with $Y \perp X, JX$; the J-plane (X,JX); and the mixed plane (X,T_λ) with $T_\lambda = \lambda^{-1/2}T$ — each expressed in terms of K_1, R_1 and $\nabla_T R_1$. Tracing, and using the pseudohermitian Bianchi identity $\sum_{\mathrm{cyc}} R(X_\alpha,X_\beta)X_\gamma = \sum_{\mathrm{cyc}} d\theta(X_\alpha,X_\beta)R_1X_\gamma$ (Tanaka), one obtains

$$\mathrm{Ricci}^L_\lambda(T_\lambda,T_\lambda) = \frac{n\lambda}{2} - \frac{1}{\lambda}\|R_1\|^2, \tag{31}$$

$$\mathrm{Ricci}^L_\lambda(X,X) = -\frac{\lambda}{2} + (R_2X,X)_1 + (2-n)(R_1X,JX)_1 - \frac{1}{\lambda}\big((\nabla_T R_1)X,X\big)_1, \tag{32}$$

for unit $X \in Q$. In particular (31) is positive iff $\lambda \geq \sqrt{2/n}\,\|R_1\| = (1/\sqrt{2n})\|\mathcal{L}_T J\|$.

Proposition 25. *If there exists a constant λ with $\mathrm{Ricci}^L(g_\lambda) > 0$, then for every unit $X \in Q$ and every $m \in M$:*

$$(R_2X,X) > \frac{1}{\sqrt{2n}}\|R_1\|, \qquad (R_2X,X) > \frac{\sqrt{2}}{1+|2-n|\sqrt{2n}}\big|((\nabla_T R_1)X,X)\big|^{1/2},$$

$$(R_2X,X) > (4n)^{-2/3}\big|\,\mathrm{Tr}_Q(\nabla\cdot R_1)(X)\big|^{2/3},$$

where $\mathrm{Tr}_Q(\nabla\cdot R_1)(X) = \sum_{i=1}^n \big((\nabla_{X_i}R_1)X,X_i\big) + \big((\nabla_{Y_i}R_1)X,Y_i\big)$ is the trace of $V \mapsto (\nabla_V R_1)X$ restricted to Q.

The proof combines (31), the symmetries $(R_2JX,JX) = (R_2X,X)$, $(R_1JX,J^2X) = -(R_1X,JX)$, $((\nabla_T R_1)JX,JX) = -((\nabla_T R_1)X,X)$, the requirement that the trinomial (32) in λ have positive discriminant, and the estimate $|\,\mathrm{Tr}(R^L_\lambda(\cdot,X)T)|^2 < \mathrm{Ricci}^L_\lambda(X,X)\,\mathrm{Ricci}^L_\lambda(T,T)$ obtained by decomposing a general tangent vector as $X + \alpha T$; finally $\mathrm{Tr}(R^L_\lambda(\cdot,X)T) = \mathrm{Tr}_Q(\nabla\cdot R_1(X))$.

Remark 26. So Riemannian Ricci positivity forces R_2 to dominate not only R_1 but also $\mathrm{Tr}_Q(\nabla R_1)$ and $\nabla_T R_1$ — the weight-3 and weight-4 quantities. In dimension ≥ 5 the contact criterion needs only R_1. **The Riemannian hypothesis is strictly more constraining.**

10 §6. Carnot–Carathéodory geodesics and a pseudohermitian Myers theorem

The CC distance between two points is the infimum of lengths of Legendrian (everywhere horizontal) paths joining them. Taking the first variation of the energy $E(\gamma) = \int_0^1 \|\dot\gamma(t)\|^2 dt$ over Legendrian variations (the constraint being $X\cdot\theta(V) = (V,JX)$, which conversely integrates to a Legendrian family with fixed endpoints) gives:

Proposition 27 (CC geodesic equation)**.** *A Legendrian curve γ with $X = \dot\gamma(t)$ is a CC geodesic iff $\nabla_X X$ is collinear with JX:*

$$\nabla_X X = -\alpha\, JX, \qquad X\cdot\alpha(t) = -(\mathrm{Tor}(T,X),X) = \tfrac12\big(J(\mathcal{L}_T J)X, X\big).$$

On H^3 this is transparent: $\pi : H^3 \to H^3/\langle T\rangle = \mathbb{R}^2$ is a Riemannian submersion and θ projects to the area form $A = \frac12(x\,dy - y\,dx)$. A closed loop in $\mathbb{R}^2$ lifts to a closed Legendrian loop iff its enclosed area vanishes; Legendrian paths joining M_1, M_2 are the lifts of curves with fixed endpoints and fixed swept area. So CC geodesics project to curves of minimal length enclosing a fixed area — **arcs of circles** — and $\nabla_X X = kJX$ with k the curvature, constant because J is T-invariant.

Second variation (dimension 3). Requiring the family γ_u to be Legendrian ($d\theta(V,X) = -c'(t) = -(V, JX)$) and constant-speed reduces the admissible variation fields along a geodesic to

$$V = c(t)\alpha(t)X + c'(t)JX + c(t)T, \qquad c(0) = c(l) = 0 = c'(0) = c'(l), \tag{33}$$

i.e. to a single free function $c(t) = \theta(V)$. A long computation, using the pseudohermitian Bianchi identity to recognize $(R(T,X)X, JX) = (\nabla_X R_1 X, JX) - (\nabla_{JX} R_1 X, X)$ and $(R(JX,X)X, JX) = -(R_2X, X)$, gives

$$\begin{aligned}\frac{\partial^2}{\partial u^2}E(\gamma_u) = \int_0^l\ & c''^2 + 2cc''(R_1X, JX) - c'^2\big((R_2X,X) + \alpha^2\big)\\ & + cc'\big(\alpha(R_1X,X) + 2(\nabla_{JX}R_1X, X) - (\nabla_X R_1X, X)\big)\\ & + c^2\big((R_1X,JX)^2 + 2\alpha^2(R_1X,JX) + (\nabla_T R_1X, X) + \alpha(\nabla_X R_1X, X)\big)\ dt.\end{aligned} \tag{35}$$

On the Heisenberg group this collapses to $\int_0^l c''^2 - c'^2\alpha^2\,dt$, which for $c(t) = 1 - \cos(2\pi t/l)$ is strictly negative when $l > 2\pi/\alpha$: **strongly curved geodesics fail to minimize even over short lengths** — exactly the area interpretation above. The term $\int -c'^2\alpha^2$ dominates on any contact manifold along strongly curved geodesics, so to control *all* geodesics one needs positivity of (R_2X, X), and indeed one wants that term to dominate all the others in (35), since the others change sign when X is replaced by JX.

Theorem 28 (pseudohermitian Myers, dimension 3)**.** *Let M^3 carry a* complete *pseudohermitian structure (equivalently, by Strichartz, geodesically complete for ∇) such that for every unit $X \in Q$,*

$$(R_2X, X) > \max\Big\{2^{10}\|R_1\|,\ 2^{16/3}\|\nabla_Q R_1\|^{2/3},\ 2^{9/2}\|\nabla_T R_1\|^{1/2}\Big\}.$$

Then M is compact, with an explicit bound $D \le D_0$ on its CC diameter, where D_0 is a universal numerical constant divided by $\min\{(R_2X,X)\cdots : X \in Q,\ \|X\| = 1\}$.

Remark 29. The displayed formula for D_0 on p. 328 of the original is garbled in the scan; read the exact constant off the printed paper. The structure is $D_0 = 4\sqrt{2}\,\pi\big/\min\{\cdots\}$ with the minimum taken over unit $X \in Q$ of a product of powers of (R_2X, X).

Proof sketch. Any two points are joined by a CC geodesic (Strichartz), so it suffices to show that every geodesic of length $> D_0$ carries a variation field V with $V^2\cdot l(\gamma) < 0$. Take V as in (33) with $c(t) = 1 - \cos(2\pi t/l)$; estimating (35) crudely gives bounds on $\partial_u^2 E$ and $\partial_u^2 l$, and the expression is negative as soon as $\frac18(2\pi^2/l^2)\big((R_2X,X) + \alpha^2\big)$ dominates each of the eight remaining terms for all α. That translates into the list of inequalities on p. 329, e.g.

$$l^2 > \frac{32\pi^2}{(R_2X,X)}, \quad 32\|R_1\| < (R_2X,X), \quad l < \frac{\pi}{4}\frac{(R_2X,X)^{1/2}}{\|R_1\|}, \quad l < \frac{\pi}{16}\frac{(R_2X,X)}{\|\nabla_Q R_1\|},\ \ldots$$

and the curvature hypotheses guarantee that an l satisfying all of them exists; one may then take $l = D_0$. □

Remark 30 (closing remark of the paper). In this geodesic approach the curvature quantities involved (R_1, $\mathrm{Tr}_Q(\nabla R_1)$, $\nabla_T R_1$) are of the *same* order as those in the Riemannian Myers theorem (by Proposition 14 above). So, unlike the cohomological criterion, the CC and Riemannian routes give essentially equivalent conditions here.

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11 What is most load-bearing for the boundary-value problem

Five points to carry forward.

1. **The normalization matters.** Rumin's Δ_Q uses degree-dependent coefficients $(n-k), (n-k+1)$, giving order 2 away from the middle and order 4 at $k = n, n+1$. Much of the later literature squares uniformly instead. Consequently a weight $x^m P$ is degree-dependent and the resulting operator is a **Douglis–Nirenberg system**, not a single operator. Fix and state the convention early.

2. **The square roots $P^{\pm}$ are worth exploiting.** In middle degree $\Delta_Q = (P^{\pm})^2$ with $P^{\pm}$ second order and self-adjoint. Building a $0H$ parametrix for a second-order object is likely far cleaner than for a fourth-order one, and §4 of Part II shows the technique is usable — with the caveat that P is signless.

3. **The Helffer–Nourrigat/Rockland criterion is the ellipticity condition at the front face.** Injectivity of $\Pi(P)$ over all nontrivial irreducible representations is exactly what the normal operator must satisfy. The comparison trick $A \gtrsim B$ of Proposition 6 is a practical device for verifying it without inverting anything.

4. **The exceptional set $|\lambda| \in \{n, n+2, n+4, \dots\}$ is where the indicial roots will live.** Proposition 2 puts Δ_H on $\Omega^{p,q}$ in the form $-\sum(X_k^2 + Y_k^2) + i(p-q)\partial_T$, and the Folland–Stein operators fail to be hypoelliptic precisely at those λ. In a boundary calculus these values are exactly the resonances/indicial roots that determine the weighted-space Fredholm thresholds. The bidegree decomposition is thus not decorative: it is how the normal operator's inversion gets organized degree by degree.

5. **The extreme bidegrees $(n, 0)$ and $(0, n)$ are the hard ones**, and for S^3 ($n = 1$) that means degree 1 — the worked example lands squarely on them.